

confirmation of the predicted relationships between computerization and task change. Occupations making relatively large increases in computer use saw relatively greater increases in labor input of nonroutine cognitive analytic and interactive tasks and larger declines in labor input of routine cognitive skills. Each of these relationships is significant at the 10 percent level or greater, and is of sizable magnitude: for all three cognitive task measures, the computerization variable more than fully accounts for the observed change in occupational task input. Only in the case of routine manual tasks, where the point estimate is close to zero, do we fail to find the expected relationship.

To examine whether intra-occupational task changes are implicitly captured by shifts in the educational and gender distribution of employees within an occupation, we add controls for the contemporaneous change in the percentage of workers in an occupation who are college graduates, high school graduates, and females.²⁴ As is visible in specifications 2 and 3, the relationship between computerization and within-occupation task change is surprisingly insensitive to these controls. In fact, standard measures of educational and gender composition are poor proxies for changes in job tasks observed by DOT examiners. In net, these findings demonstrate that shifts in job content away from routine tasks and toward nonroutine cognitive tasks are a pervasive feature of the data and are concentrated in industries and occupations that adopted computer technology most rapidly.

VI. QUANTIFYING THE MAGNITUDE OF TASK STRUCTURE CHANGES

What is the economic significance of the change in the tasks performed by the U. S. labor force during the last three decades? The answer is not immediately apparent since units of task input do not have a familiar scale. To quantify task shifts in concrete economic terms, we draw together task changes within industries, education groups, and occupations to calculate their potential contribution to the demand for college-educated labor during 1970 to 1998. This analysis proceeds in three steps.

We begin by estimating a "fixed coefficients" model of educa-

24. For consistency of measurement, we employ CPS computerization, education, and gender means by occupation for 1984 to 1997. We cannot perform an analogous exercise using the NIPA investment measures since they are not available for occupations.

tion requirements in industries and occupations as a function of their task inputs:

$$(17) \quad \text{College Share}_j = \alpha + \sum_{k=1}^4 \pi_k \cdot T_j^k + \varepsilon_j.$$

In this equation, College Share_j is the college-equivalent share of employment (in FTEs) in industry or occupation j at the midpoint of our samples, and the T_j^k 's measure industry or occupation task input in centiles during the same period.²⁵ The coefficients, $\hat{\pi}_k$, obtained from (17) provide an estimate of college-equivalent labor demand as a function of industry or occupation task inputs. We estimate this model separately for industry and occupation task demands, using employment data from our CPS samples from 1980 and 1984 paired to the 1977 DOT job task measures.²⁶

We refer to equation (17) as a fixed coefficients model because it neglects the impact of task prices on task demands—or equivalently, assumes that the elasticity of substitution between college and noncollege equivalent workers is zero. This is an imperfect approximation: if the market price of nonroutine relative to routine tasks has risen, this calculation will understate demand shifts favoring nonroutine tasks and, by implication, college graduate employment.

The second step of our methodology is to translate task shifts into predicted changes in college employment. We first assemble changes in our four key task measures over 1970 to 1998, $\Delta T_{1970-1998}^k$. We then apply these task shift measures to the “fixed coefficients” estimated from equation (17) to calculate

$$(18) \quad \tilde{\Delta} \text{College Share}_{1970-1998} = \sum_{k=1}^4 \tilde{\pi}_k \cdot \Delta T_{1970-1998}^k.$$

Here, $\tilde{\Delta} \text{College Share}_{1970-1998}$ is the change in the college share of aggregate employment predicted by task shifts over 1970–1998.

25. We follow Autor, Katz, and Krueger [1998] and Murphy, Romer, and Riddell [1998] in defining college equivalent workers as all those with a college degree or greater plus half of those with some college. Results using exclusively college graduates are quite similar.

26. The industry task demand model is estimated using the 1980 MORG employment data, which is at the midpoint of our sample. The occupation task demand model is estimated using the 1984 CPS sample, which is at the midpoint of our occupation sample. For completeness, estimates of equation (17) also control for input of nonroutine manual tasks. Inclusion or exclusion of this covariate has no substantive impact.

The intuition for this calculation is that industries and occupations that are high in nonroutine cognitive task input, and low in routine manual and routine cognitive task input, employ college graduates relatively intensively. Consequently, secular increase in nonroutine cognitive task input and declines in nonroutine cognitive and manual task input over 1970–1998 will cause equation (18) to predict corresponding growth in the college graduate share of aggregate employment.

Table VII summarizes the changes in job task input due to cross-occupation (extensive margin) and within-occupation (intensive margin) shifts documented by our earlier analyses. Panel A presents observed economywide shifts in task input during the period 1970 to 1998. Panel B presents analogous numbers where in place of observed task changes, we tabulate changes in task input predicted by computerization. Specifically, we use estimates of equations (14)–(16), corresponding to the models in Tables IV–VI, to calculate the predicted mean change in each task measure due to contemporaneous industry or occupation computerization. A limitation of this approach is that it treats computerization as an exogenous determinant of industry and occupation task change. Since, as stressed above, we view computer adoption and task change as simultaneously determined, we view this exercise as primarily illustrative.

We implement these calculations in panels C and D. Panel C uses equation (18) to estimate the extent to which rising input of nonroutine tasks and declining input of routine tasks raised the college share of aggregate employment over 1970–1998. As seen in columns 1–4, observed cross-occupation (extensive margin) task changes raised college employment by 2.1 percentage points per decade between 1970 and 1998. Three-quarters of this contribution (1.5 percentage points) is due to shifts favoring nonroutine cognitive tasks. The remainder is explained by shifts against routine cognitive and manual tasks.

Columns 5–7 perform analogous calculations for 1980 to 1998. Here we add within-occupation (intensive) margin task change for 1977 to 1991. In net, shifts favoring nonroutine over routine tasks contributed 2.5 percentage points growth per decade to college-equivalent employment over these eighteen years.²⁷

27. Observed intensive margin shifts did not contribute to this demand growth, however, due to the offsetting effects of routine cognitive and nonroutine analytic tasks. This stands in contrast to within-occupation task changes predicted by computerization, where intensive margin shifts are economically large.

TABLE VII
SHIFTS IN COLLEGE-EQUIVALENT LABOR DEMAND IMPLIED
BY CHANGES IN JOB TASKS, 1970–1998

	1. 1970– 1980 extensive margin	2. 1980– 1990 extensive margin	3. 1990– 1998 extensive margin	4. 1970– 1998 extensive margin	5. 1980– 1998 extensive margin	6. 1980– 1998 intensive margin	7. 1980– 1998 extensive + intensive
A. $10 \times$ observed annual changes in DOT task measures (percentile changes relative to 1960 task distribution)							
Nonroutine analytic	3.02	2.97	3.12	3.04	3.05	−0.39	2.67
Nonroutine interactive	4.68	5.31	4.48	4.84	4.85	0.58	5.43
Routine cognitive	−0.14	−3.48	−4.88	−3.03	−4.26	−2.76	−7.02
Routine manual	1.63	−1.47	−3.88	−1.44	−2.81	0.74	−2.07
B. $10 \times$ predicted annual changes in DOT task measures (percentile changes relative to 1960 task distribution)							
	NIPA computer input measure				CPS computer use measure		
Nonroutine analytic	0.84	1.35	2.30	1.55	2.56	0.54	3.10
Nonroutine interactive	1.47	2.36	4.01	2.70	3.16	1.04	4.20
Routine cognitive	−1.05	−1.68	−2.86	−1.92	−3.14	−3.32	−6.46
Routine manual	−1.15	−1.86	−3.15	−2.12	−2.84	0.32	−2.52
C. $10 \times$ predicted annual changes in college-equivalent share of employment in percentage points, due to observed task shifts (panel A)							
Nonroutine tasks	1.53	1.40	1.63	1.51	1.53	−0.36	0.83
Routine tasks	−0.20	0.66	1.17	0.50	0.97	0.27	0.87
All tasks	1.33	2.06	2.80	2.01	2.49	−0.09	2.40

Panel A: Observed extensive margin task shifts are defined as the change in economywide input of each task (in percentiles of the 1960 task distribution) estimated using DOT 1977 occupational task measures applied to Census and CPS samples for 1970 to 1998 and summarized in Table II. Intensive margin shifts are measured as change in mean of DOT occupational task input (measured in centiles of the 1977 task distribution) between 1977 and 1991 DOT revisions, using the 1980 and 1998 occupational distributions of employment from the CPS MORG samples. See Table I and Appendix 1 for definitions and examples of task variables.

Panel B: Predicted task changes are calculated as the weighted mean of the NIPA computer investment measure or CPS computer use measure (as noted) multiplied by the coefficient from a regression of changes in industry or occupation task input on the relevant computer measure. NIPA coefficient estimates correspond to specification (1) of Table IV. CPS extensive margin computer task estimates correspond to Panel A of Table V. CPS intensive margin computer task estimates correspond to specification (1) of Table VI. (A negligible interaction term is ignored.)